INCH-POUND

MIL-PRF-1/1619F
9 August 2016
SUPERSEDING
MIL-PRF-1/1619E
8 April 2016

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPES 8161 AND 8251

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein
shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triodes, ceramic, metal.
See figures 1 and 2.
Mounting position: Vertical, base down or up.
Weight: 8161: 6.25 pounds nominal.
8251: 7.5 pounds nominal.

ABSOLUTE RATINGS: C telegraphy

Type 8161:

Parameter:	F1	Ef	Eb	Ec	Ib	Pg	Pp	Pi	Anode core and seal T	Cooling
Unit:	MHz	V ac	kV dc	kV dc	A dc	W	kW	kW	°C	
Maximum	110	7.9	4.0	-1.0	2.0	150	4.0	7.2	250	---
Minimum	---	7.1	---	---	---	---	---	---	---	---
Test conditions:	---	7.5	3.0	Adj.	0.83	---	---	---	---	1/

ABSOLUTE RATINGS: C telegraphy

Type 8251:

Parameter:	F1	Ef	Eb	Ec	Ib	Pg	Pp	Pi	Anode core and seal T	Cooling
Unit:	MHz	V ac	kV dc	kV dc	A dc	W	kW	kW	°C	
Maximum	30	7.9	4.0	-1.0	2.0	150	4.0	7.2	250	---
Minimum	---	7.1	---	---	---	---	---	---	---	---
Test conditions:	2/									

See footnotes at end of table I.

GENERAL:
Qualification - Required.
This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.
Service-life guarantee - 500 hours shall be required for tube operating time.
Marking - Tubes sold under service-life guarantee shall be marked with contract number and with the number of operating (500 hours) guaranteed.

AMSC N/A

FSC 5960

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TABLE I. Testing and inspection.

Inspection	Method MIL-STD-1311	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1							
Filament current	1301			If	49	54	A ac
Peak emission	1231		eb = ec = 2,500 v	is	35	---	a
Electrode voltages (grid)	1261			Ec	-67	-100	V dc
Total grid current	1266	5/		Ic	---	-50	μA dc
Primary grid emission	1266		Ef = 8.3 V ac; t = 15 sec; Ic = 0.9 A dc; anode grounded	Isg	---	-250	μA dc
Conformance inspection, part 2							
Amplification factor	1316A			Mu	19	26	---
Direct interelectrode capacitance	1331			Cgp Cin Cout	16.8 29.2 0.60	23.2 40.2 1.20	pF pF pF
Power oscillation	1236		Eb = 4,000 V dc; Ib = 1.8 A dc; F = 110 MHz; Rg = 2,150 ohms	Po	4.2	---	kW useful

See footnotes on top of next page.

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TABLE I. Testing and inspection - Continued.

1/ In all electrical tests involving application of filament voltage, forced-air cooling is allowable as follows: base-to-anode airflow, 67 cfm maximum; anode-to-base airflow, 84 cfm maximum through the anode cooler, plus 3 cfm maximum directed into the filament stem structure. These figures apply for air at 40°C maximum and at sea level. At higher air temperature or altitude, suitable correction of the airflow is allowable. Cooling air should be supplied before or simultaneously with application of electrode voltages (including the filament) and may be removed simultaneously with them.

2/ The 8251 type may be made by attaching the base flying leads to an 8161 which has met all the requirements listed herein. After attachment of the flying leads and any other mechanical processing, including plating, the 8251 shall meet the requirements of the holding period and the total grid current test.

3/ For the cooling test, the tube shall be mounted in an infinite baffle system, or equivalent, so that all air passing in a base-to-anode direction shall pass through the anode cooler of the tube. With cooling air of not less than 20°C nor more than 40°C, both base and anode shall be cooled by applying an airflow of 67 cfm maximum at sea level. At altitudes above sea level, the equivalent mass flow may be used. An appropriate value of anode voltage and current shall be used to produce the specified anode dissipation. Temperatures shall be measured by means of thermocouples, located as follows:

Anode core: The thermocouple shall be embedded in the base of the anode core, at the base of the cooling fins, by means of drilling a small hole shallow enough so that the tube vacuum shall not be lost, placing the welded thermocouple junction therein, and then peening the edges of the hole to hold the thermocouple firmly in place.

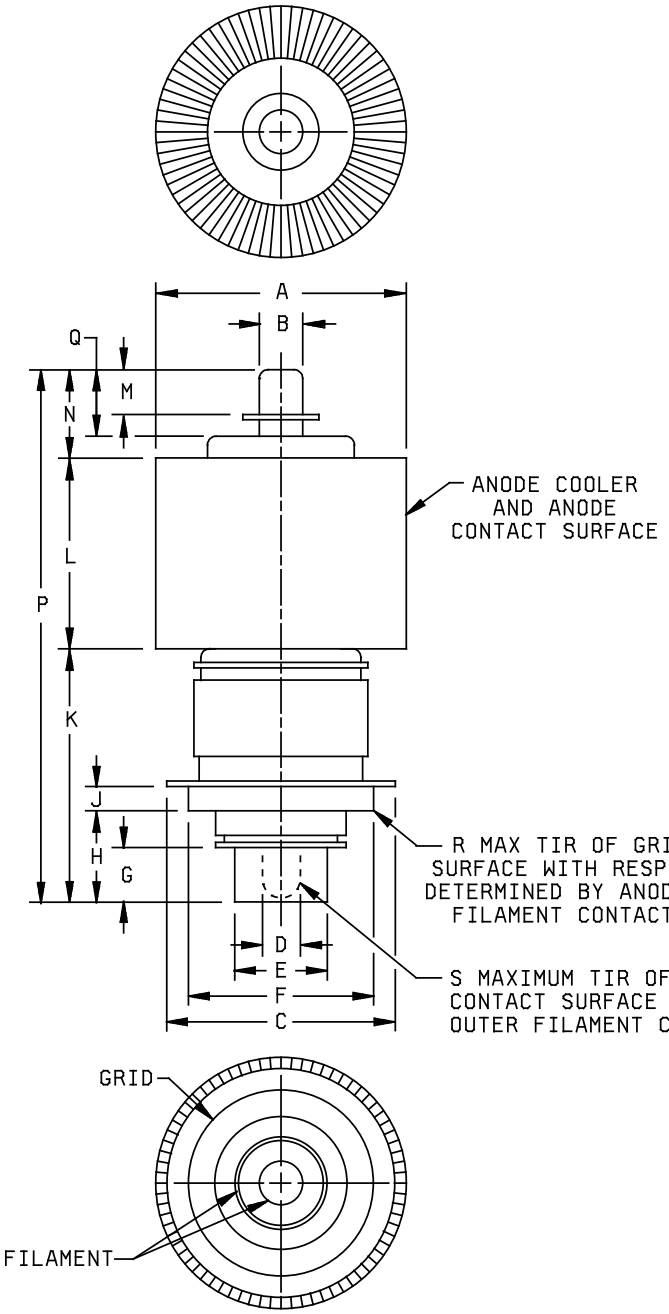
Filament seal: The thermocouple shall be attached using a suitable cement to either the inner or the outer filament conductor and as close as possible to where the two conductors join the stem dielectric material.

Good electrical continuity between the thermocouple and the metal surface to which it is attached shall be demonstrated before the cooling test is performed. The size and material of the thermocouples, their installation, and the measuring instrument used to determine temperature shall be in accordance with good engineering practice.

4/ The pressure drop across the anode cooler shall be measured while the tube is operating as specified during the cooling test. If test is performed at altitudes significantly above sea level and the airflow is adjusted for an equivalent mass-flow rate, the allowable pressure drop may also be recomputed on the same basis.

5/ This test to be performed first at the conclusion of the holding period.

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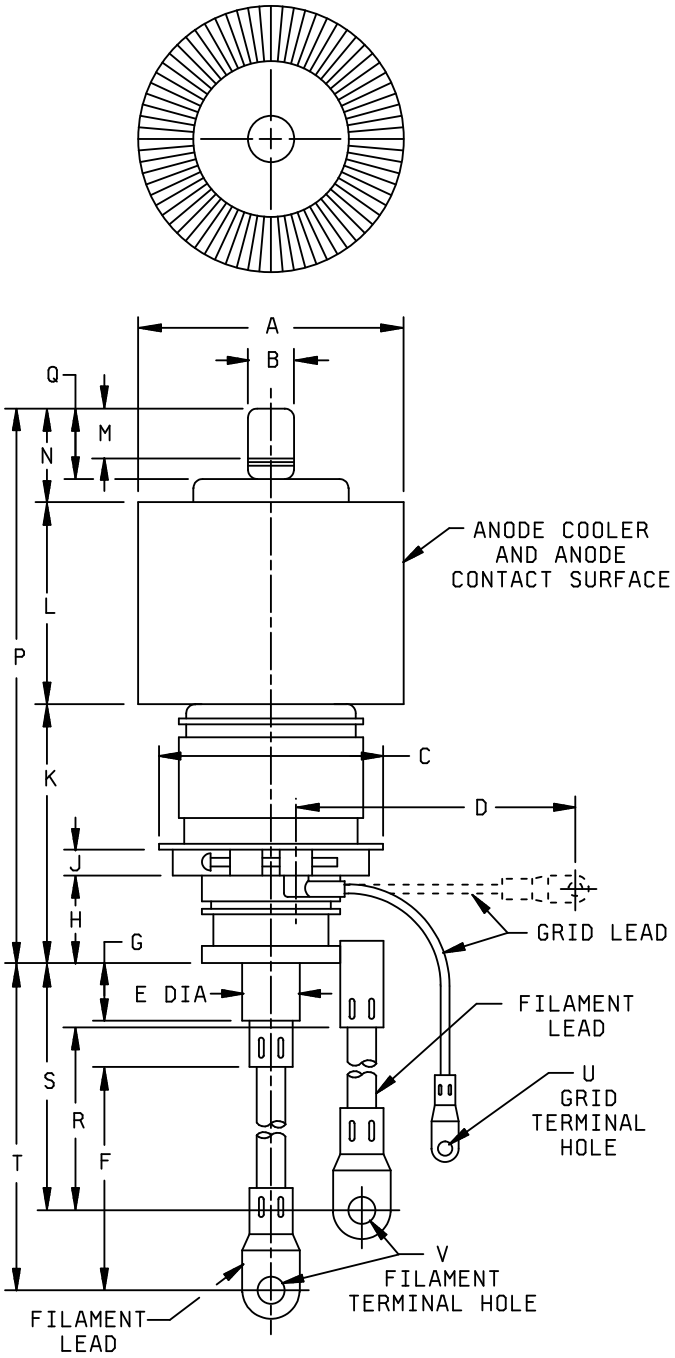


Ltr	Dimensions (see note 1)			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	4.094	4.156	103.99	105.56
C	---	3.625	---	92.08
D	.615	.635	15.62	16.13
E	1.490	1.510	37.85	38.35
F	2.990	3.010	75.95	76.45
G	.812	.938	20.62	23.83
H	1.375	1.625	34.93	41.28
P	8.000	9.000	203.20	228.60
R	---	.040	---	1.02
S	---	.025	---	0.64
Conformance inspection, part 3				
B	.781	.844	19.84	21.44
J	.375	.438	9.53	11.13
K	3.875	4.250	98.43	107.95
L	2.937	3.063	74.60	77.80
M	.688	.812	17.48	20.62
N	1.188	1.688	30.18	42.88
Q	1.000	1.125	25.40	28.58

NOTES:
1. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 8161

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Ltr	Dimensions (see note 1)			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	4.094	4.156	103.99	105.56
C	---	3.625	---	92.08
D	6.375	6.625	161.93	168.28
G	.812	.937	20.62	23.82
H	1.375	1.625	34.93	41.28
P	8.000	9.000	203.20	228.60
S	7.937	8.437	201.60	214.30
T	8.937	9.437	237.00	239.70
U	.194	.200	4.93	5.08
V	.385	.395	9.78	10.03
Conformance inspection, part 3				
B	.781	.844	19.84	21.44
E	.859	.890	21.82	22.61
F	7.000	7.500	177.80	190.50
J	.375	.438	9.53	11.13
K	3.875	4.250	98.43	107.95
L	2.937	3.063	74.60	77.80
M	.688	.812	17.48	20.62
N	1.188	1.688	30.18	42.88
Q	1.000	1.125	25.40	28.58
R	7.000	7.500	177.80	190.50

NOTES:
1. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Outline drawing of electron tube type 8251.

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Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

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Navy - EC
Air Force - 85
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(Project 5960-2016-028)

Review activities:
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